THABANG MOLISE

Software Engineer — Backend Security & Robotics Systems

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PROFILE

Final-year Software Engineering student at Belgium Campus iTversity, focused on backend systems and the hardware they connect to. On my final-year project, I co-led an eight-person team building the BC Smart Ecosystem, a campus navigation and drone food-delivery platform that is now deployed. My responsibilities covered feature delivery, backend security and the drone hardware. I also lead the campus Robotics Club, where I teach members controllers, wiring and CAD.

TECHNICAL SKILLS

Languages

- Python
- Java
- C#
- JavaScript / TypeScript
- SQL

Backend & Data

- Node.js & Express
- REST API Design
- PostgreSQL & Prisma
- MongoDB & Mongoose
- Socket.IO

Security

- JWT & Session Management
- bcrypt Password Hashing
- Role-Based Access Control
- Stripe Payment Integration
- SQL-Injection Prevention
- Fault Isolation Between Services

Engineering Practice

- Git / GitHub Branching & PR Review
- CI Pipelines
- Test Definition
- Technical Documentation
- Agile / Kanban

Hardware & Edge AI

- Arduino & Raspberry Pi 5
- PWM Servo Control & Driver HATs
- Power Regulation
- Computer Vision (YOLOv8, OpenCV)
- Monocular Depth (MiDaS)

Frontend & Tools

- React
- Flutter
- Postman
- Appium
- Fusion 360 (Foundational)

TEAM PROJECTS

BC Smart Ecosystem — Traceback-Technology

Node.js, Express, MongoDB, Stripe, Python

Co-Group Leader · Backend Security & Hardware Lead | PRJ381, 8-person team | Deployed, in active development

- Co-led an eight-person team building BC Ways (campus navigation) and BC Eats (food ordering and drone delivery), deployed at tracebacktechnologies.co.za.
- Owned feature delivery: checked each feature was built and shipped to requirements, and tracked scope creep across all squads.
- Acted as backend lead, assigning work, running feasibility checks project-wide and closing security gaps such as missing session management.
- Built the backend security layer: Stripe payment logic, SQL-injection protection, and module isolation so a failure in BC Ways cannot take down BC Eats, or the reverse.
- Led the drone hardware track: researched drone mechanics and architectures, and wrote the hardware proposal documents used to approach potential sponsors.
- Designed the delivery demo: a DJI Neo flown by software through Appium-automated controls, delivering to a custom carrier with its own telemetry, ETA and release system.

CivicConnect — Software Engineering Project (SEN381)

GitHub Lead · Backend Security | 3-person team

- Owned all GitHub work: chose and approved the branching strategy, defined the CI checks and pipelines, and reviewed, approved or rejected pull requests.
- Set the repository's documentation structure, separate from the formal project documentation.
- Implemented the security features, including role-based access control (RBAC), and defined the project's tests.

ROBOTICS & AI PROJECTS

6-Axis Robotic Arm with Vision Pipeline — Robotics Showcase

Python | YOLOv8, MiDaS, OpenCV | Arduino / Raspberry Pi

- Built a 6-axis servo arm driven by an Arduino Uno or Raspberry Pi 5 over PWM, using a buck converter to step 12V down to 5V and protect the servos.

- Combined YOLOv8 object detection with MiDaS monocular depth estimation and a calibrated camera to locate objects in front of the arm.
- Delivered a working showcase in four days, using a scripted motion sequence for the live demo.

Autonomous Garbage Collector (AGC) — Personal Project

Python | TensorFlow, OpenCV | Raspberry Pi

In development since 2025

- Designing a self-driving vehicle that detects and collects litter: six-wheel omnidirectional drive, a robotic arm, an onboard bin and a sensor suite.
- Built object-detection demo models and 3D-printable arm designs; development runs in three stages, from proof of concept to a deployment-ready unit.

PERSONAL PROJECTS

MAEA (Market Analysis Execution Agent)

Python | MetaTrader 5 | Reinforcement Learning

In development

- Building an autonomous agent that analyses markets, places trades and edits its own analysis configuration within governance rules set by its owner.
- Learning design: populations of evolution-strategy agents share a replay buffer, sampled by a central reinforcement-learning agent that models how the market reacts.
- Optimises for regime resilience and population diversity; agents are retired on portfolio depletion, sustained regression or stagnation.
- Run as a formal software engineering project, with every proposed change documented against agreed acceptance criteria.
- Evolved from an earlier **Trading Bot** for EURUSD/GBPUSD: LightGBM models on a multi-timeframe strategy (H4 bias, M15 entries), regime detection, a deterministic risk engine and a high-impact news filter.

Classical Tracker (Vista Etude)

Flutter, Java | Prisma

- Built a local-first piano practice app with a sheet-music viewer and auto-scroll, practice-session recording and reference-recording storage.
- Designed the backend: Prisma schema, REST endpoints and file upload and streaming, with optional cloud backup and no accounts, to match its single-user scope.

LEADERSHIP

Head of Robotics Club — Belgium Campus iTversity, Pretoria

2025 – present

- Designed beginner robotics projects to bring new members into the field, and organised and ran the club's sessions and workshops.
- Taught members microcontrollers, wiring and programming their first controllers, plus foundational CAD for designing 6-axis robotic arms.
- Presented past students' robotics projects at university open days, and developed new projects for future showcases.
- Supported final-year students on their robotics projects, helping them connect and program their hardware.
- Wrote semester reports reviewing the club's challenges and proposing structured projects, hackathons and intercampus collaboration.

EDUCATION

Bachelor of Computing (Software Engineering) — Final Year

Belgium Campus iTversity, Pretoria, 2024 – present

Project modules: PRJ381 (final-year project), SEN381 (software engineering)

National Senior Certificate

St. Andrew's School, Welkom — 2023

LANGUAGES

English (Proficient) | Sesotho (Native) | Afrikaans (Intermediate) | Japanese (Elementary)